

AUDIO5822.

Description.

The AUDIO5822 is a digital audio amplifier board based on the TAS5822 Class-D audio amplifier. The board provides a hardware platform for digital audio processing, amplification, and evaluation through an I²S audio interface and I²C control interface.

Features.

- TAS5822 Class-D audio amplifier.
- PVDD(Power Supply): 4.5V to 26.3V.
- DVDD(Digital supply): 1.8V or 3.3V.
- Audio Output: Mono.
- I²S digital audio interface.
- I²C control interface.
- Integrated digital audio processing.
- High-efficiency Class-D architecture.
- Digital volume control.
- Programmable audio configuration.
- High efficiency reverse-polarity power protection.
- ESP32-compatible HDR-F interface
- 5.00 mm pitch PVDD and speaker output connectors.

Applications.

- ESP32-based audio systems.
- Bluetooth audio receivers.
- Portable audio equipment.
- Smart speakers and IoT audio devices.

Pins.

- +V(PVDD)
- +3.3V(DVDD)
- ADR/FAULT
- LRCLK
- SCLK
- SDIN
- SDOUT
- SDA
- SCL
- PDN
- GND
- OUT+
- OUT-

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Hardware Design.

The AUDIO5822 board integrates the TAS5822 digital audio amplifier with a control and data interface, input power protection, local supply decoupling, bootstrap components, and a differential LC output filter. The hardware is organized into functional blocks to simplify integration, testing, and maintenance. The board design is inspired by TI's recommended design in the TAS5822 datasheet for MONO (PBTL) mode; see Section 8.2.2, page 74.

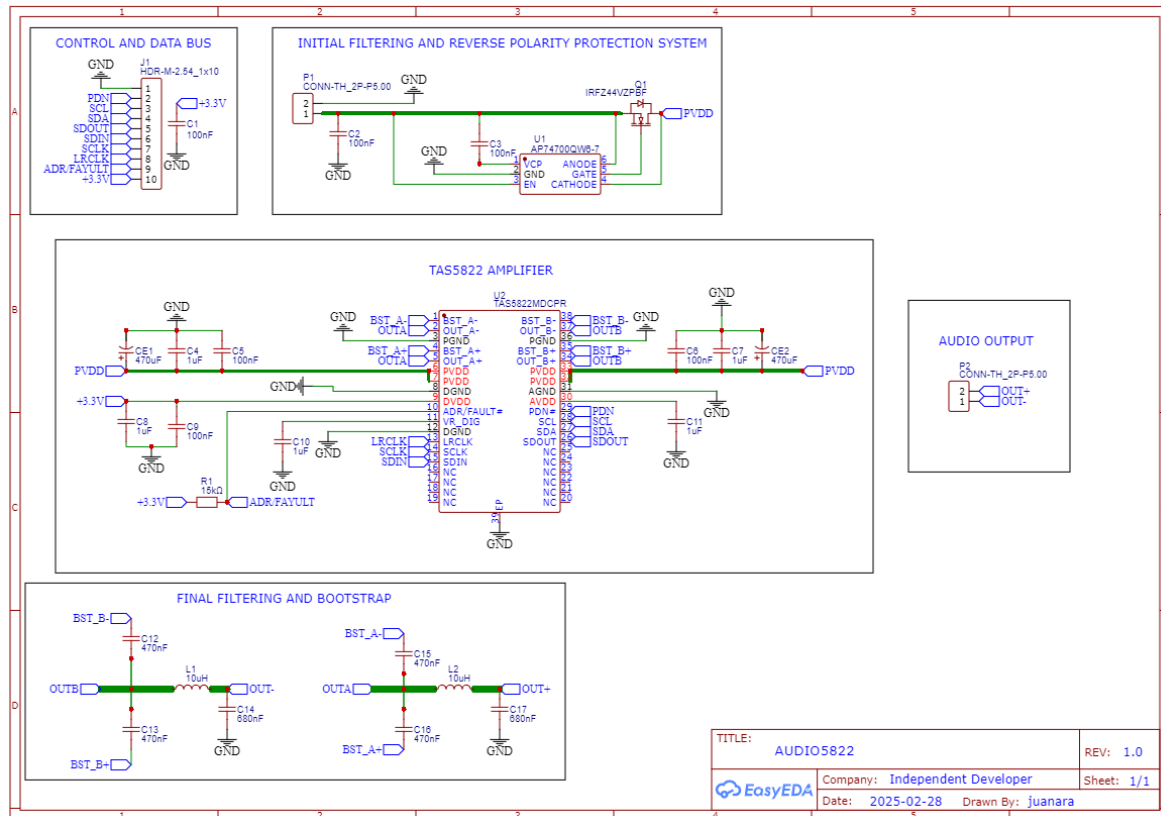


Figure 1. AUDIO5822 schematic.

Control and data bus.

The AUDIO5822 board provides a digital interface for connection to an external host controller. The interface includes the I²C control bus, I²S digital audio bus, power-down control, and the ADR/FAULT signal. The connector is primarily intended for integration with ESP32-family controllers, but the interface can also be used with other 1.8-V or 3.3-V digital systems, provided that the corresponding logic-supply and pull-up requirements are satisfied. C1 (100 nF) provides high-frequency decoupling for the +3.3 V (DVDD) rail and should be placed as close as possible to the +3.3 V connector.

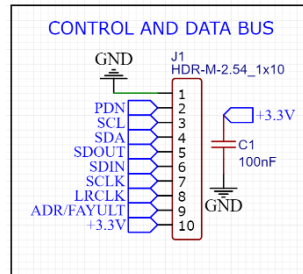


Figure 2. Control and data bus.

Signal	Function
+3.3V(DVDD)	Digital interface supply provided by the board.
ADR/FAULT	Multipurpose signal used to select the I ² C address and, when configured, report fault conditions.
LRCLK	I ² S word-select/left-right clock defining the audio frame and channel boundaries.
SCLK	I ² S bit clock defining the timing of the serial audio data.
SDIN	Serial audio data input carrying the digital audio stream to the TAS5822.
SDOUT	Serial audio data output from the TAS5822. It can provide digital audio data for monitoring or other processing functions.
SDA	I ² C serial data line used to configure and control the TAS5822.
SCL	I ² C serial clock line used for communication with the TAS5822.
PDN	Power-down control input. Allows the host controller to place the amplifier in its power-down state.
GND	Ground reference for the digital interface.

Initial filtering and reverse polarity protection system.

The AUDIO5822 power input incorporates an initial filtering and reverse-polarity protection stage. The circuit receives the external PVDD supply through P1 and uses the AP74700QW6-7 controller together with a power MOSFET (IRFZ44VZPBF) to prevent damage caused by an incorrectly connected supply. Local capacitors provide input filtering and high-frequency decoupling before the protected PVDD rail is distributed to the amplifier stage.

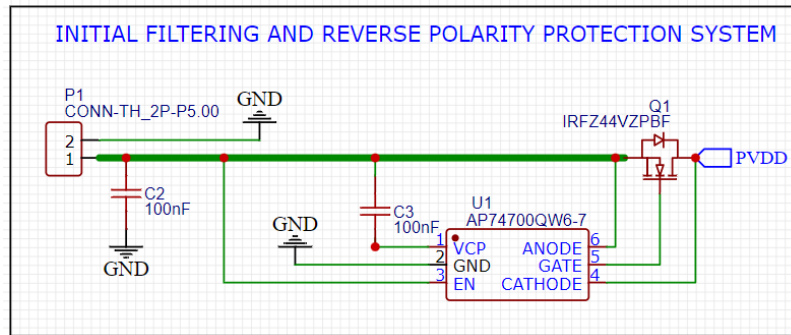


Figure 3. Initial filtering and reverse polarity protection.

Component	Value/Part	Function
P1	5.00 mm terminal	PVDD power input
U1	AP74700QW6-7	Reverse-polarity protection controller
Q1	IRFZ44VZPBF	Power MOSFET used as protection element
C2	100 nF	Input filtering / decoupling
C3	100nF	Charge-pump capacitor

TAS5822 amplifier.

The TAS5822 is the central device of the AUDIO5822, integrating the digital audio processing and power amplification stages required to drive a speaker load. The device is powered through the PVDD supply for its power stage and through a separate +3.3 V(DVDD) supply for its digital circuitry. The PVDD rail is supported by a local decoupling network consisting of bulk and high-frequency capacitors. CE1 (470 μ F) provides bulk energy storage, while C4 (1 μ F) and C5 (100 nF) provide local decoupling across progressively higher frequency ranges. A similar network is implemented on the opposite side of the device using CE2 (470 μ F), C7 (1 μ F), and C6 (100 nF). The digital supply is decoupled with C8 (1 μ F) and C9 (100 nF), providing a low-impedance local supply for the device's digital circuitry.

Audio data is transferred to the TAS5822 through its digital I²S interface. The board exposes the LRCLK, SCLK, and SDIN signals, allowing the amplifier to interface with a wide range of microcontrollers or digital audio processors capable of providing a compatible I²S interface. The I²S interface is dedicated to audio-data transmission, while device configuration and monitoring are handled independently through the I²C control interface.

The I²C interface consists of the SCL and SDA signals and provides access to the internal TAS5822 registers for device configuration, operating-mode control, and monitoring. The ADR/FAULT pin provides the address-selection and fault-related functionality of the device. On the AUDIO5822, R1 (15 k Ω) connects this pin to the +3.3V(DVDD) supply, establishing the required logic level for the selected operating configuration. The PDN input is also exposed to allow the external controller to control the power-down state of the amplifier as part of the system initialization and shutdown sequence.

The TAS5822 provides four power-output terminals: OUT_A+, OUT_A-, OUT_B+, and OUT_B-. In BTL operation, these terminals form two independent differential output channels. In the AUDIO5822, the device is configured for MONO (PBTL) operation, in which the output stages are operated in parallel to increase the available output power for a single loudspeaker load. The external output network provides separate filtering paths for the positive and negative sides of the final differential mono output. Because the loudspeaker is driven differentially, neither speaker terminal should be connected to GND or to another amplifier output.

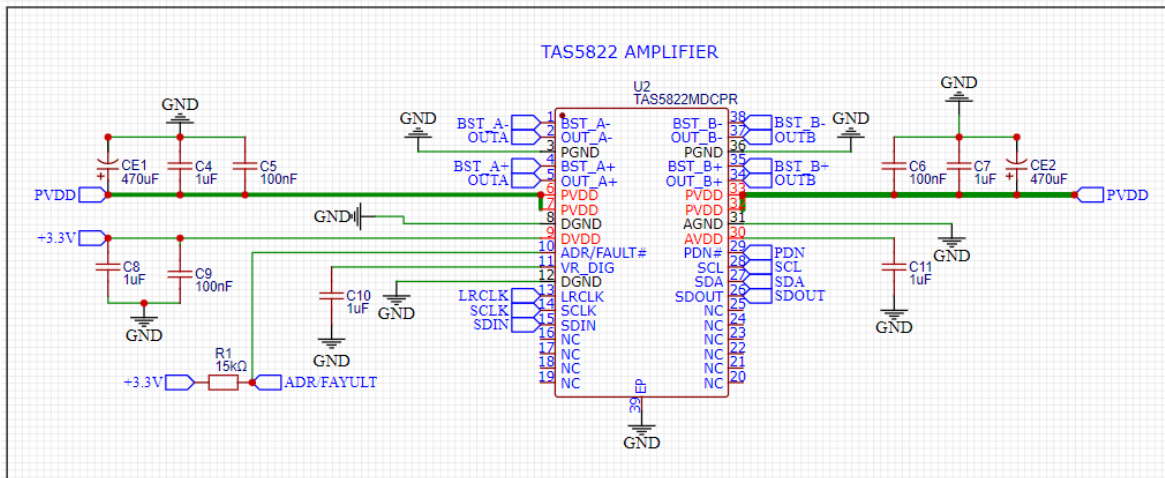


Figure 4. TAS5822 Amplifier.

Component	Value/Part	Function
U2	TAS5822MDCPR	Processes the digital audio input and drives the parallel output stages used for the differential mono speaker output.
CE1, CE2	470 μ F	Bulk energy-storage capacitor for the PVDD supply. Helps provide transient current to the amplifier and reduces low-frequency supply variations.
C4 and C7	1 μ F	Local PVDD decoupling capacitor for the amplifier power stage, reducing supply impedance over a wider frequency range.
C5 and C6	100 nF	High-frequency PVDD decoupling capacitor located close to the power-stage connections to suppress high-frequency supply noise.
C8	1 μ F	Local decoupling capacitor for the +3.3 V digital supply. Helps maintain a stable supply for the TAS5822 digital circuitry.
C9	100nF	High-frequency decoupling capacitor for the +3.3 V supply. Reduces high-frequency noise and transient disturbances on the digital supply.
C11	1 μ F	Local filtering capacitor associated with the auxiliary/control supply circuitry. Helps stabilize the corresponding supply node and reduce voltage fluctuations.
C10	1 μ F	Capacitor connected to the VREG pin. Provides the required local filtering and stabilization for the TAS5822 internal regulator.
R1	15 k Ω	Pull-up resistor connected to the ADR/FAULT pin. Establishes the required logic level for the device's address-selection configuration and provides the appropriate bias for the fault-reporting function.

Final filtering and bootstrap.

The final filtering and bootstrap stage provides the interface between the TAS5822 switching output stage and the external loudspeaker. It combines two essential functions: bootstrap supply generation for the high-side gate drivers and LC low-pass filtering of the Class-D switching waveform.

In the AUDIO5822, the TAS5822 is configured for mono operation using Parallel BTL (PBTL). In this configuration, the output stages are operated in parallel to drive a single loudspeaker load. The resulting mono differential output is routed through the output LC filter before reaching the speaker connector.

Bootstrap Network

The TAS5822 uses high-side NMOS output devices that require a floating gate-drive supply. This supply is generated by the internal bootstrap circuitry and the external bootstrap capacitors connected to the corresponding bootstrap pins.

During the switching operation, the bootstrap capacitors are charged when their associated output nodes are driven low. The stored charge then provides the floating voltage required to drive the high-side NMOS devices.

The AUDIO5822 implements four dedicated 470 nF bootstrap capacitors, following the recommended component value for the TAS5822:

Component	Value	Function
C12	470nF	Bootstrap capacitor for OUT_B-
C13	470nF	Bootstrap capacitor for OUT_B+
C15	470nF	Bootstrap capacitor for OUT_A-
C16	470nF	Bootstrap capacitor for OUT_A+

LC Filter.

The TAS5822 is a Class-D amplifier and therefore generates a high-frequency switching waveform at its power outputs. The output LC network attenuates the high-frequency switching components while allowing the desired audio-frequency signal to reach the loudspeaker.

For the AUDIO5822, the output filter uses 10 μ H series inductors and 680 nF shunt capacitors. This configuration provides low-pass filtering of the differential mono output and reduces the high-frequency switching energy present at the speaker terminals.

The selected 10 μ H + 0.68 μ F LC configuration follows the recommended values specified by Texas Instruments for the applicable TAS5822 operating conditions.

Reference	Value	Function
L1	10uH	Filter LC
C14	680nF	Filter LC
L2	10uH	Filter LC
C17	680nF	Filter LC

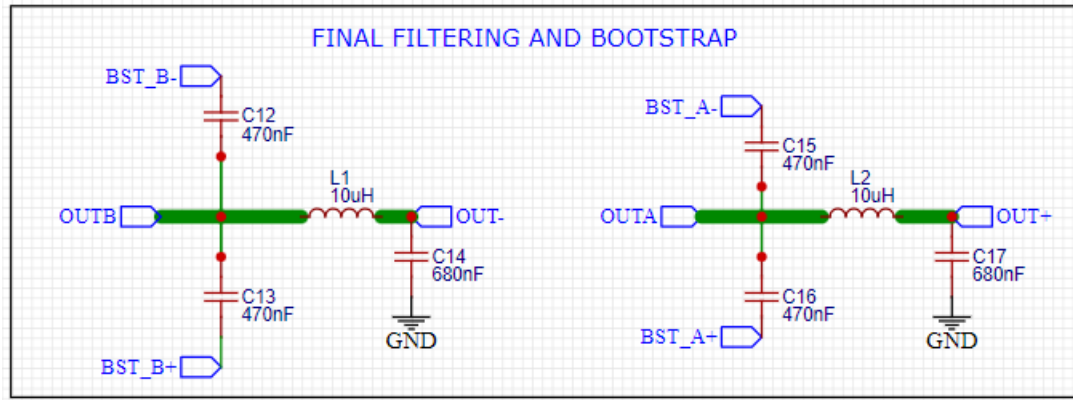


Figure 5. Final filtering and bootstrap.

Audio Output.

The Audio output block provides the amplified differential audio signal from the AUDIO5822 to the external speaker. The connection is implemented through P3, a 2-pin, 5.00 mm pitch terminal connector.

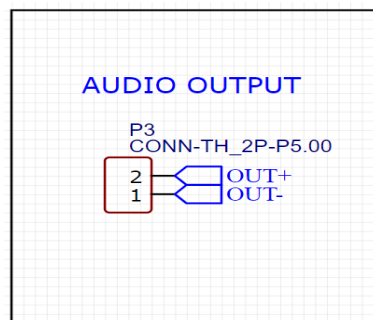


Figure 6. Audio Output.

PCB Layout.

The AUDIO5822 PCB layout was developed with emphasis on proper power distribution, short high-current paths, signal integrity, and effective separation between the digital control interface and the power amplification section. The placement and routing follow the design recommendations provided by Texas Instruments for the TAS5822, particularly for operation in MONO (PBTl) mode.

The AUDIO5822 is implemented on a 4-layer PCB with overall dimensions of 80.01 mm × 56.007 mm. The board uses a standard green solder mask and is organized into dedicated layers for signal routing, ground, and power distribution. Figures 7 and 8 show the top and bottom views of the AUDIO5822 PCB, respectively.

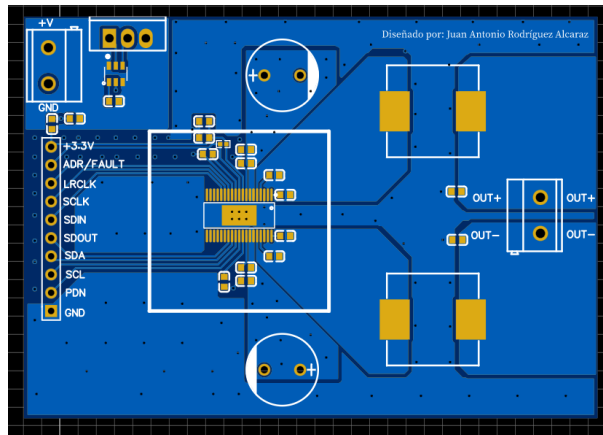


Figure 7. Top Side(2D).

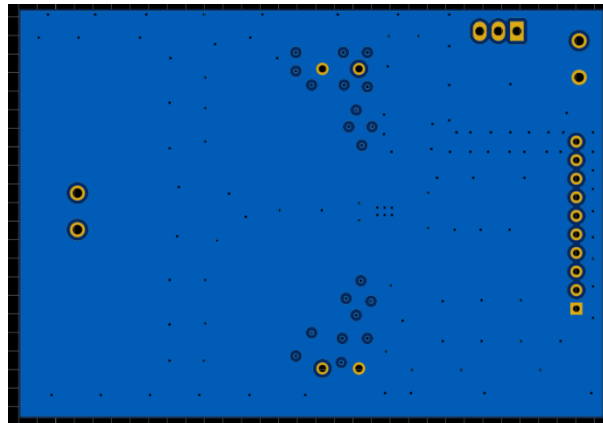


Figure 8. Bottom Side(2D).

Hardware Recommendations.

For future implementations or revisions of the AUDIO5822 PCB, several hardware improvements may be considered to facilitate integration, testing, and debugging.

I²C pull-up resistors: It is recommended to include dedicated pull-up resistors for the SDA and SCL lines on the PCB. This would provide the required biasing for the I²C bus and allow the board to operate more independently from the external controller.

Dedicated test points: Additional test points are recommended for the main power, control, and digital audio signals. Providing accessible measurement points for PVDD, DVDD, GND, PDN, SDA, SCL, LRCLK, SCLK, and SDIN would simplify debugging, commissioning, and laboratory measurements.

Power and status indication: A dedicated power or status indicator could be considered to provide a quick visual indication of the board's operating state during development and testing.

Add dedicated mechanical mounting points: The PCB could be improved by incorporating dedicated mounting holes at the board corners or other mechanically suitable locations. These mounting points would allow the AUDIO5822 to be securely fastened to an enclosure, chassis, or mechanical support, improving mechanical stability and simplifying system integration. The mounting holes should maintain adequate clearance from copper traces, planes, vias, and other components. If metallic screws or standoffs are intended to be used, the mounting-hole design should also consider electrical isolation and grounding requirements.

Comments.

The AUDIO5822 was designed as a modular audio amplification platform that can be integrated into different embedded audio systems.

One of the initial applications is an ESP32-based Bluetooth audio receiver. In this application, the ESP32 receives audio wirelessly and transmits the digital audio stream to the AUDIO5822 through the I²S interface, while the TAS5822M performs the digital Class-D amplification and drives the external loudspeaker.

Another application under development is an ESP32-S3-based SD audio player. In this system, audio files stored on an SD card are processed by the ESP32-S3 and transmitted digitally to the AUDIO5822 through the I²S interface.

These applications illustrate the modular architecture of the platform, where the external microcontroller is responsible for the audio source and digital processing, while the AUDIO5822 provides the amplification stage and speaker interface.

The separation between the digital controller and the power amplification stage allows the same AUDIO5822 hardware platform to be reused with different embedded systems and audio sources.